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Social Trust and Reciprocity Among Adolescent Girls Exposed to Interpersonal Violence

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Abstract

Trauma research has traditionally focused on altered emotion regulation and its role in psychopathology, whereas mechanisms of social behavior remain comparatively unexplored, particularly among adolescents. It has been previously reported that adolescents with histories of interpersonal violence (IV) demonstrate disrupted social learning, and the degree to which they are impaired during social interactions requiring trustful behaviors may be associated with their levels of anxiety. In the present study, 52 adolescent females (n = 26 control; n = 26 IV-exposed) between ages of 11 and 17 completed a multi-round adaptation of the Trust Game in which they interacted with a confederate peer run by a computer program, alternating between the roles of investor and investee. The task was designed to operationalize the social behaviors of trust and trust reciprocity, where the magnitude of the participants' monetary investment in the confederate during the investor role represented trust while the proportion of investment returned to the confederate in the investee role represented trust reciprocity. IV-exposed and control participants did not differ in trust (i.e., as investors); however, IV-exposed participants without anxiety

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diagnoses demonstrated lower trust than those with anxiety diagnoses. For trust reciprocity (i.e., as investees), there were again no differences between IV-exposed participants and controls; however, IV-exposed participants with anxiety diagnoses had increased trust reciprocity compared with both other groups. Similarly, caregiver-reported anxiety symptoms were associated with trust reciprocity behaviors among the IV-exposed adolescents. Findings suggest that IV exposure and associated anxiety impacts adolescents' trust behaviors, demonstrating potential mechanisms for maladaptive social behavior among trauma-exposed youth.

Keywords
interpersonal violence, PTSD, anxiety, trust, reciprocity

Introduction

Exposure to interpersonal violence (IV) during early childhood increases the risk of developing psychopathology and maladaptive behaviors during adolescence. Rates of reported IV, including physical or sexual abuse, among children and adolescents in the United States are around 12.5%, and estimates including unreported cases are around 41% (Sumner et al., 2015). Although post-traumatic stress disorder (PTSD) is the hallmark post-traumatic diagnosis, with around 25% of adolescents developing PTSD following IV (Alisic et al., 2014), anxiety, mood, and substance use disorders are also prevalent and often persist into adulthood (Auxéméry, 2018; Kilpatrick et al., 2003; McLaughlin et al., 2010). Although the majority of research investigating the robust link between IV and psychopathology focus on emotional processes such as emotion regulation (Chang, Kaczkurkin, McLean, & Foa, 2018; Ehring & Quack, 2010; Heleniak, Jenness, Vander Stoep, McCauley, & McLaughlin, 2016; Villalta, Smith, Hickin, & Stringaris, 2018), less research has been aimed at examining social processes that might be impacted by early life IV.

Key aspects of social behavior include trust, that is, believing in the reliability of others, and trust reciprocity, the act of returning trust, thereby creating stable two-way relationships (Bellucci, Chernyak, Goodyear, Eickhoff, & Krueger, 2017). Trust, reciprocity, and cooperation are prevalent in our social lives, serving as keystones to building meaningful, healthy relationships. Neuroeconomic theories assert that individuals weigh their own beliefs regarding the likelihood of return—or in this case the reciprocation of trust—when deciding to invest (Camerer, 2003; Fehr & Camerer, 2007). Consequently,

trust and cooperation require an initial assessment of risk prior to making an investment. A common laboratory model of social trust and reciprocity is an economic exchange paradigm referred to as the Trust Game (Berg, Dickhaut, & McCabe, 1995). The multi-round adaptation of the Trust Game measures trust and trust reciprocity behaviorally through a two-player economic exchange with players alternating between investing and receiving monetary units in participation with a confederate (Belli, Rogers, & Lau, 2012). Within the task, interpersonal trust is operationalized by the dollar amount the participant in the "investor" role decides to give, or invest, in the "investee." The investment multiplies by a factor of three before the investee receives a dollar amount. Trust reciprocity is operationalized by the proportion of monetary units the investee returns to the investor (Anderl et al., 2018). Within each round of the Trust Game, cooperation occurs when the investor and investee both engage in generous giving behavior. Continual cooperation between both participants is necessary to maximize mutual benefit (Belli et al., 2012; King-Casas et al., 2008).

The Trust Game has been used to conceptualize changes in trust and reciprocity in adults with past trauma exposure and psychopathology (Bell, Robinson, Katona, Fett, & Shergill, 2019; Gobin & Freyd, 2014; Williams et al., 2014). In the case of interpersonal trauma, the process of risk assessment in trust behaviors may be compromised; past victimization influences one's ability to accurately attend to risk and make good decisions regarding one's own safety (Walsh, DiLillo, & Messman-Moore, 2012). Adults with histories of childhood sexual abuse and high betrayal trauma exhibit reduced rates of trust, and lower levels of general trust were associated with reduced amounts of investment in the Trust Game (Gobin & Freyd, 2014). These results were corroborated by Bell et al. (2019) who found that adults with IV-related PTSD exhibited lower investments with cooperative partners than controls; however, they did not differ from controls when interacting with unfair partners. In addition, male combat veterans who completed long-term PTSD treatment exhibited improvements in their investment ratios compared with those who received short-term treatment or no treatment (Williams et al., 2014). Other studies have also found dysfunctional trust behaviors among adults with social anxiety or borderline personality disorder (Anderl et al., 2018; King-Casas et al., 2008). Socially anxious individuals in particular had been found to have decreased trust reciprocity, indicating anxiety may also play a role in trust (Anderl et al., 2018).

Although the Trust Game has been explored among IV-exposed adult populations, it has not yet been implemented in an IV-exposed adolescent

population. It is unclear whether adolescents might exhibit similar behaviors to adults, or if the alterations develop later in life. Adolescence is a key time for the development of prosocial behaviors (Güroğlu, van den Bos, & Crone, 2014; Tamnes et al., 2018), and notably, trust behaviors continue to develop throughout adolescence (Lemmers-Jansen, Krabbendam, Veltman, & Fett, 2017). Among healthy adolescents, individuals with a greater aptitude for perspective-taking—an increased ability to acknowledge the perspectives of others—tend to extend more trustful behavior in times of cooperation (Fett et al., 2014). When cooperation is broken, these same individuals tend to engage in less reciprocal behavior in response to their peer's unfair behavior. IV-exposed girls exhibited weaker responses to trust violations, or unexpected negative social outcomes, during a social learning task, suggesting altered social learning of trust (Lenow, Scott Steele, Smitherman, Kilts, & Cisler, 2014). The effects on trust behaviors may be compounded by the presence of affective psychopathology, including depression or anxiety, which have also been shown to lead to poorer outcomes in the ability to engage in prosocial behaviors such as trust and trust reciprocity (DiLillo & Long, 1999; Feiring, Simon, Cleland, & Barrett, 2013; van Rijsewijk, Dijkstra, Pattiselanno, Steglich, & Veenstra, 2016). In a longitudinal study among individuals with or without histories of childhood maltreatment, worse peer relationships and overall relationship functioning predicted lower resilience against psychopathology in adulthood (Collishaw et al., 2007). However, this investigation did not control for adolescent psychopathology, and the deficits in social functioning may be the result of psychological symptoms manifesting in adolescence that lead to worse outcomes in adulthood. Further investigations are needed on how the impact of IV and subsequent psychopathology influences trust throughout development.

To our knowledge, social trust and reciprocity have not been directly examined among youth exposed to early life IV. Based on the well-established risk for psychopathology following trauma (Alisic et al., 2014; Auxéméry, 2018; McLaughlin et al., 2010), and evidence suggesting that behavioral trust and reciprocity are disrupted among individuals with psychopathology (Anderl et al., 2018; King-Casas et al., 2008; Williams et al., 2014), we hypothesized participants with histories of IV exposure would have impaired social trust and trust reciprocity relative to healthy control participants, indicated by lower investments and reciprocation. Given the high prevalence of anxiety symptoms following IV (Auxéméry, 2018; Collishaw et al., 2007), we also hypothesized that the degree of social impairment would be positively correlated to the degree of anxiety or PTSD symptoms among trauma-exposed youth.

Method

Participants and Assessment

Fifty-two adolescent girls, aged 11 to 17 (n = 26 controls) and their legal guardians enrolled in the study following approval from the local Institutional Review Board. Inclusion criteria for the IV-exposed group required a history of direct assault exposure (i.e., physical abuse by a caregiver, physical assault resulting in injuries or life threat, and sexual assault, abuse, or molestation) reported during the National Survey of Adolescents (NSA) trauma assessment. Participants were excluded if they presented psychotic symptoms, a primary substance use disorder, pervasive developmental disorder, or major medical disorder. Control participants were included based on female sex and age, and excluded based on history of trauma or mental health disorder. IV-exposed participants were not excluded if they took psychiatric medications, requiring medication stability for at least 4 weeks. Clinical and demographic characteristics of the adolescent sample are provided in Table 1. Caregivers were also recruited to complete the Child Behavior Checklist (CBCL) regarding the adolescent participant. The only inclusion criteria required of caregivers was legal guardianship of the adolescent participant.

Study Measures

Multi-round adaptation of the trust game. Participants played a multi-round adaptation of the trust game (Belli et al., 2012; Cisler et al., 2015; King-Casas et al., 2008; van den Bos, van Dijk, Westenberg, Rombouts, & Crone, 2011), and were informed that they were playing the task with another adolescent girl. In reality, participants were interacting with a computer program. Participants and the computerized confederate alternated between the role of "investor" and "investee" (see Figure 1). Participants completed three blocks as "investee" with the computerized "investor" and three blocks as "investor" with the preprogrammed "investee," counterbalancing role order across participants. Investors started with US$10 at the beginning of each round, followed by an offer to share any proportion of the US$10 with investees. Any chosen amount tripled before the investees received investments. Investees could then choose to return any proportion of the investment and keep the remaining balance. A computerized confederate was used to selectively choose the proportion of monetary units invested or returned. Computerized confederate investors gave participant investees US$4 to US$6 at Block I (Rounds 1-5), US$1 to US$3 at Block II (Rounds 6-10), and US$4 to US$6 at Block III (Rounds 11-15). Confederate investees reciprocated participant investors' with 40% to 60% of their investment at Block I, 10% to 30% at

Table 1. Clinical and Demographic Characteristics of the Sample.

	Control (a) N = 26		Assaulted N = 26		
Variable N = 52	*M*	*SD*	*M*	*SD*	*p* Values
Age	16.34	1.64	15.66	1.86	.169
IQ	120.38	16.20	105.42	18.67	.003
Ethnicity					.459
White	60.0%		57.7%		
Black	12.0%		19.2%		
Asian	8.0%		0.0%		
Hispanic, Latina	4.0%		11.5%		
Other	16.0%		11.5%		
Direct assaults	0		3.62	2.25	*p <.001*
Time since last assault	N/A		1.69	2.67	
Sexual assault	0.0%		76.9%		*p <.001*
Physical assault	0.0%		76.9%		*p <.001*
Anxiety disorder					
Panic disorder	0.0%		19.2%		.021
Agoraphobia	0.0%		53.8%		*p* < .001
Separation anxiety	0.0%		7.7%		.157
Social anxiety	0.0%		19.2%		.021
Specific phobia	0.0%		11.5%		.080
Generalized anxiety	0.0%		30.8%		.003
OCD	0.0%		15.4%		.041
PTSD	0.0%		53.8%		*p* < .001
Mood disorder	0.0%		53.8%		*p* < .001
CAPS total severity	0.19	0.98	40.92	31.19	*p* < .001
UCLA PTSD-RI	1.85	3.60	29.35	16.02	*p* < .001
CBCL anxiety	2.42	2.32	8.12	6.35	*p* < .001
CBCL depression	1.46	1.86	4.69	3.58	*p* < .001
Medication					
Antidepressants	0.0%		15.4%		.041
SSRI	0.0%		34.6%		.001
SNRI	0.0%		3.8%		.322
Benzo	0.0%		3.8%		.322
Mood Stabilizers	0.0%		7.7%		.157
Stimulants	0.0%		11.5%		.086
Other	0.0%		7.7%		.157

Note. OCD = Obsessive-Compulsive Disorder; PTSD = post-traumatic stress disorder; CAPS = clinician-administered PTSD scale for *DSM* IV; UCLA PTSD-RI = University of California at Los Angeles Post-Traumatic Stress Disorder Reaction Index, a self-report of PTSD symptoms for adolescents; CBCL = child behavior checklist, scores from the anxiety and depression subscales reported by the parent; MINI-KID = Mini-International Neuropsychiatric Interview for Children and Adolescents; SSRI = selective serotonin reuptake inhibitor; SNRI = serotonin and norepinephrine reuptake inhibitor; *DSM* IV = Diagnostic and Statistical Manual of Mental Disorders (4th ed).

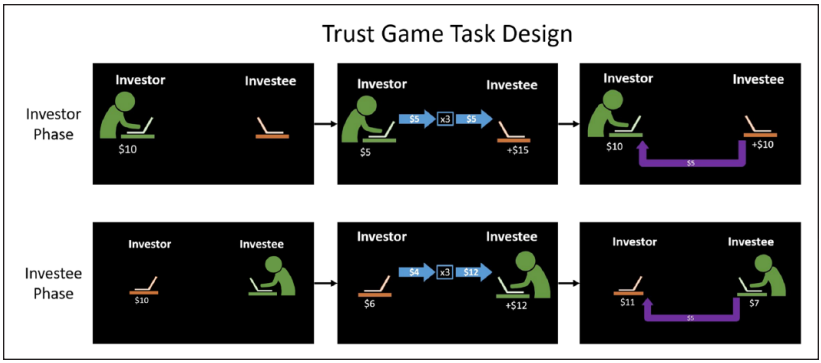

Figure 1. Task paradigm for both the investor and investee roles.
Note. (Top) Participant acts as the investor and begins with US$10. The participant chooses a certain amount of money to give to the confederate investee. Whatever amount the participant chooses to invest will triple in value before reaching the investee. The investee then chooses how much of their money to give back to the investor. (Bottom) Task paradigm where the participant acts as the investee and the confederate as the investor.

Block II, and 40% to 60% again at Block III. Deception ensured that participants' behavior during the trust game would emulate participants' social trust with real adolescent peers. Prior to debriefing about the confederate at a follow-up visit, participants were asked if they believed they were playing with another adolescent participant. Four participants were excluded at this stage because they did not believe the deception, in addition to one subject who did not attend the follow-up debriefing session; thus, their belief in the deception is unknown.

NSA. The trauma portion of the NSA, a structured interview used in prior epidemiological studies of comprehensive IV exposure and mental health functioning among adolescents (Kilpatrick et al., 2000), was used to assess adolescent participants' history and chronology of childhood adversity and IV exposure. Behaviorally specific, dichotomous questions were used to assess specific assaultive events including, but not limited to the following: physical assault (threatened or actual use of a weapon, choking, assault with fists or other body parts), sexual assault (inappropriate touching, penetration, object insertion, oral sex), and severe physical abuse perpetrated by caregivers (physical discipline resulting in severe injury, choking, burning or scalding, threatened or actual use of a weapon).

The Mini-International Neuropsychiatric Interview for Children and Adolescents (MINI-KID). The MINI-KID (Sheehan et al., 2010) is a widely used diagnostic tool among individuals aged 6 to 17. Under the supervision of a licensed clinical psychologist, a trained clinical interviewer assessed participants' current and lifetime psychological functioning. The following anxiety disorders were assessed using the MINI-KID: obsessive-compulsive disorder, generalized anxiety disorder, panic disorder, agoraphobia, separation anxiety disorder, or social anxiety. Participants were grouped into three groups based on their history of IV exposure and anxiety diagnoses into the following groups: healthy control participants ($n = 26$), IV-exposed participants without anxiety diagnoses (IV^{ANX-}; $n = 7$), and IV-exposed participants with anxiety diagnoses (IV^{ANX+}; $n = 19$).

Clinician-administered PTSD Scale—Child version for the diagnostic and statistical manual of mental disorders (4th ed) (DSM-IV) (CAPS-IV). The CAPS-IV consists of 33 clinician-administered items that assess clusters of PTSD symptoms (reexperiencing, avoidance and numbing, and hyperarousal) and global severity and functioning both in the individual's lifetime and the past month (Blake et al., 1995). Individual symptoms were assessed on 5-point Likert-type scales for the frequency (rated from 0 = "none of the time" to 4 = "most of the time/almost daily") and intensity (rated from 0 = "not a problem" to 4 = "a big problem, have to stop activities") over the duration of either the past month or worst month. PTSD diagnoses were operationalized as six or more symptoms surpassing threshold (frequency ≥ 1 and intensity ≥ 2), with at least one reexperiencing symptom, three avoidance and numbing symptoms, and two hyperarousal symptoms. This is consistent with prior studies among pediatric PTSD samples (Herringa, 2017).

University of California at Los Angeles Post-Traumatic Stress Disorder Reaction Index (UCLA PTSD-RI) for the DSM-IV. This 22-item self-report measure assesses PTSD symptom severity (Steinberg, Brymer, Decker, & Pynoos, 2004). Participants indicated how much of the time in the past month they experienced each symptom on a 5-point Likert-type scale (rated from 0 = "none of the time/never" to 4 = "most of the time/almost every day"). Prior studies demonstrate strong psychometric properties of this scale (Steinberg et al., 2013).

CBCL. This 113-item caregiver-reported measure assesses the legal guardians' perception of their child's general psychological functioning (Achenbach, 1991). For each item, caregivers report how true a statement of a particular domain describes their child on a 3-point Likert-type scale

(0 = "not true," 1 = "somewhat or sometimes true," 2 = "very true or often true"). The following eight domains are assessed: aggressive behavior, anxious/depressed, attention problems, rule-breaking behavior, somatic complaints, social problems, thought problems, and withdrawn/depressed. Analyses for this manuscript focused only on the anxiety domain. The CBCL school-age version is widely used among youth aged 6 to 18 years.

Verbal IQ. Previous reports have implicated Verbal IQ to be associated with trauma history characteristics and PTSD symptoms in adolescents (Saltzman, Weems, & Carrion, 2006). Subsequently, verbal IQ was assessed and included in data analysis as a covariate to isolate the effects of IV exposure on trust behaviors. The 190-item Receptive One Word Picture Vocabulary Test (Brownell, 2000) assesses verbal IQ through a clinician-administered test. Participants are asked to match a word spoken by the interviewer to an illustration based on their understanding of the word's meaning. Item difficulty increases incrementally with multiple starting points for developmental age. The test is administered using a critical range which requires identifying the participant's basal and ceiling, the outer limits on the test that the participant correctly identifies.

Data Analysis

Participant behavior during the investor and investee roles of the Trust Game were compared between groups using linear mixed-effects models in Matlab via the fitlme.m function. Separate models were conducted for the investor and investee roles of the task. For the investor role analyses, the outcome measure was amount invested (US$0-US$10) across the 15 trials. For the investee role, the outcome measure was percent of investment returned across the 15 trials. The models consisted of a group (i.e., control vs. IV-exposed) by block (i.e., confederate cooperation during the task) interaction term with covariates for site, age, and verbal IQ, and block was estimated as a random effect nested with a random subject variable. Verbal IQ and age were included as covariates to ensure the differences in trust were not better explained by intelligence or neurocognitive development. The participant group variable of primary interest was a dummy coded variable representing control participants (n = 26) and IV-exposed participants (n = 26). A power analysis using G*Power 3.1.9.2 indicated a sample size of n = 26 per group would provide power of .8 to detect a large effect size (Cohen's d = .8). We tested the hypothesis that anxious symptoms among IV-exposed youth were related to trust and reciprocity in two ways. First, we tested the impact of anxiety diagnoses, though this analysis was underpowered (power = .5 to detect large effect

size). Second, we also tested whether continuous caregiver-rated anxiety (i.e., the CBCL anxiety subscale) and the total PTSD symptom severity from the CAPS were related to trust and reciprocity among the IV-exposed participants, which provided greater power to detect a large effect size (power = .82 to detect large effect). These models were identical to the primary group analyses, besides using Symptom Severity × Block Interaction terms with the continuous CBCL anxiety subscale or CAPS total score measures.

Results

Group Difference in Trust

Main effect of task and group differences. The block manipulation of confederate cooperation demonstrated a robust impact on participant trust behavior in the investor role, $t(773) = -5.809$, $p < .001$, and there was no significant main effect between the control and IV-exposed group, $t(773) = -0.046$, $p = .963$. The impact of block on trust behavior was not better explained by interactions with the control and IV-exposed groups $t(773) = 0.539$, $p = .590$.

Effect of anxiety diagnoses. There was no main effect between IV^{ANX+} and control participants, $t(771) = -0.943$, $p = .346$, or between IV^{ANX-} and control participants $t(771) = -1.754$, $p = .0799$. There was a significant main effect difference between IV^{ANX+} and IV^{ANX-} participants, $t(771) = -2.316$, $p = .021$, where IV^{ANX-} exhibited lower investment behavior than IV^{ANX+} (Figure 2a). There were no interactions with the block manipulation of confederate cooperation.

Effect of anxiety and PTSD symptoms among IV-exposed group. There was neither a main effect, $t(383) = 1.130$, $p = .259$, nor interactive effect, $t(383) = 1.078$, $p = .282$, of anxiety symptoms (i.e., caregiver-rated CBCL anxiety subscale) on trust. In addition, there was no main effect of PTSD symptom severity on trust, $t(383) = -0.688$, $p = .492$, nor an interaction with block, $t(383) = 1.126$, $p = .261$. These results remained insignificant when both anxiety symptoms and PTSD symptoms were included in the model together (p values > .16).

Group Difference in Trust Reciprocity

Main effect of task and group differences. In the investee role, there was no main effect of block manipulation of confederate cooperation on trust reciprocity, $t(773) = 0.474$, $p = .636$, and no interactions with the control versus IV-exposed $t(773) = -0.538$, $p = .591$. There was no significant main effect for the control versus assault groups comparison, $t(773) = 1.560$, $p = .119$.

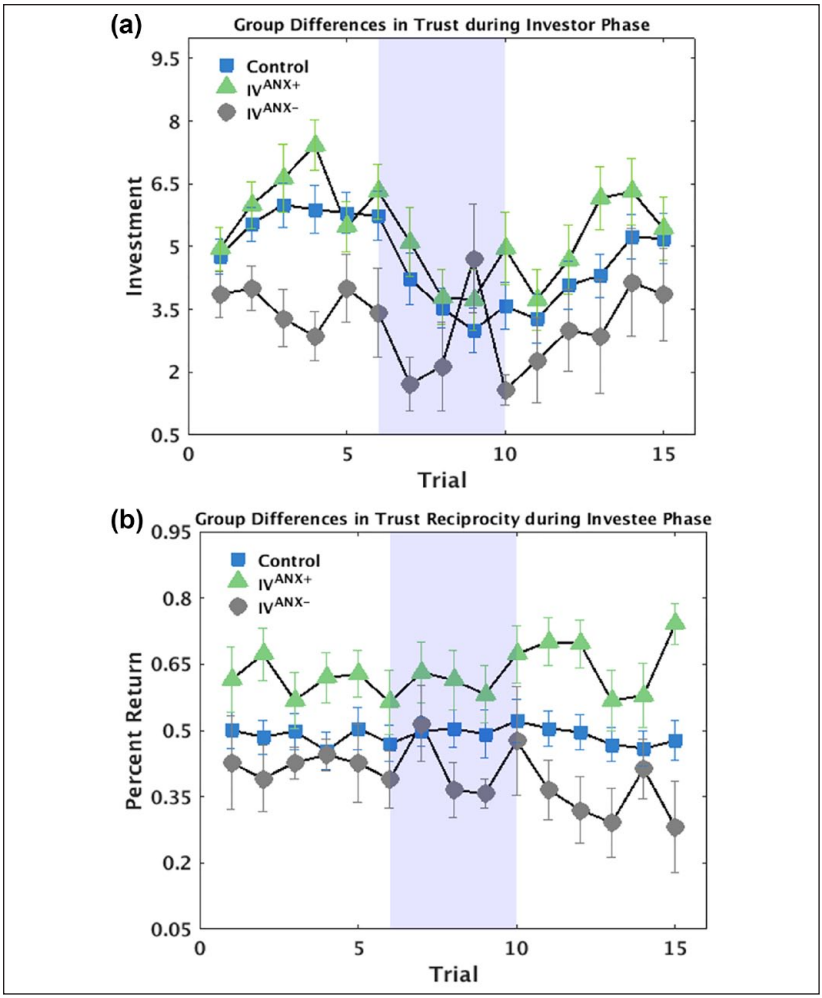

Figure 2. (a) Comparisons of trust behavior by group differences between healthy controls (blue) and adolescents with a history of IV-exposure (green) when participants were acting as the investor. Shaded region indicates Block II, where the proportion returned from the confederate to the investor decreased from US$4 to US$6 to US$1 to US$3. (b) Comparisons of reciprocity behavior when participants were acting as the investee by group differences healthy controls (blue), adolescents with a history of IV-exposure and anxiety diagnoses (IV^{ANX+}, green), and adolescents with a history of IV-exposure without anxiety diagnoses (IV^{ANX-}, gray). Shaded region indicates Block II, where the proportion invested by the confederate decreased from US$4 to US$6 to US$1 to US$3. IV = interpersonal violence, ANX = presence of anxiety diagnosis.

Effect of anxiety diagnoses. When examining whether anxiety is related to trust reciprocity, there was a significant main effect of IV^{ANX+} participants compared with controls, $t(771) = 2.770$, $p = .006$, and compared with IV^{ANX-}, $t(771) = 2.982$, $p = .003$, indicating greater reciprocity among the IV^{ANX+} (Figure 2b).

Effect of anxiety and PTSD symptoms among IV-exposed group. There was a significant main effect of caregiver-rated anxiety symptoms, $t(383) = 2.152$, $p = .0320$, not qualified by an interaction of block x anxiety rating, $t(383) = -1.742$, $p = .0823$, indicating generally greater trust reciprocity among IV-exposed girls with higher caregiver-rated anxiety symptoms (Figure 3a). There was no main effect of PTSD symptom severity, $t(383) = -0.224$, $p = .823$, nor an interaction with block, $t(383) = -0.350$, $p = .727$ (Figure 3b). The main effect of anxiety symptoms held when PTSD symptoms were included in the same model, $t(381) = 2.440$, $p = .015$, indicating that symptoms of PTSD are not the driving factor in the IV-exposed girls' trust reciprocity.

Discussion

This study is the first to investigate the relationship between IV exposure, anxious psychopathology, and trust behaviors in an adolescent population using the multi-round version of the Trust Game. The analysis identified distinct differences in trust behaviors among anxious IV-exposed adolescents in both their levels of generosity and reciprocation. As investors, participants exhibited a robust influence of confederate cooperation in their behavior; both the IV-exposed and control groups decreased their amount of monetary investments when their generosity was not reciprocated by the confederate investee. This remains consistent with various iterations of the Trust Game across both patient and healthy populations and indicates that trust changed across all participants when cooperation was ruptured (Anderl et al., 2018; Fett et al., 2014; King-Casas et al., 2008). IV-exposed participants did not differ from healthy controls during the investment role, suggesting no group differences in behavioral trust based on IV exposure alone. However, among the IV-exposed adolescents, the IV^{ANX+} group displayed increased trust compared with IV^{ANX-} group. This effect was not corroborated when examining the caregiver-reported anxiety, although the directionality of the relationship remained the same. Anxiety was found to be influential in the investee phase as well; although there was no main effect of cooperation on participant behavior, IV^{ANX+} participants exhibited increased trust reciprocity compared with IV^{ANX-} and control participants, and higher caregiver-reported anxiety levels were associated with increased trust reciprocity. Finally, PTSD

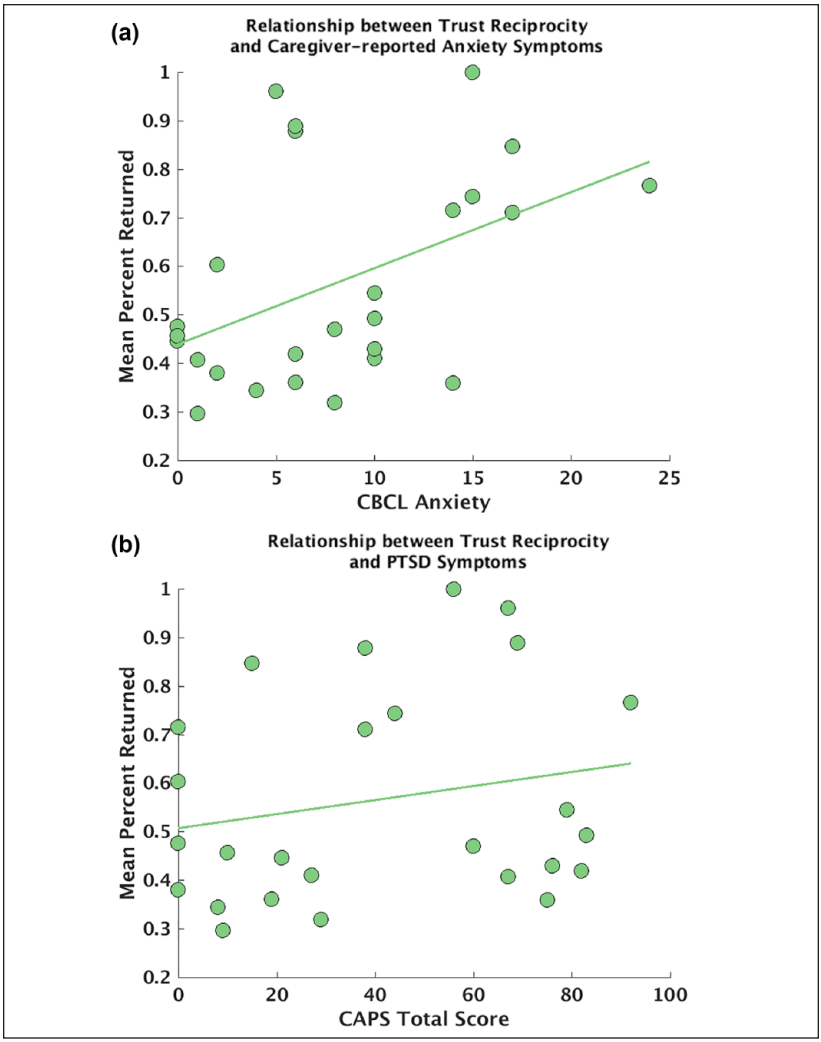

Figure 3. (a) Correlation of mean percent of investments returned and the caregiver-rated anxiety from the CBCL anxiety subscale among the IV-exposed participants. IV-exposed adolescents with greater caregiver-rated anxiety symptoms had generally greater trust reciprocity, $t(383) = 2.152$, $p = .032$. CBCL = Child Behavior Checklist. (b) Correlation of mean percent of investments returned and the total CAPS score among the IV-exposed participants. There was no main effect of PTSD symptom severity on trust reciprocity behavior, $t(383) = -0.224$, $p = .823$. CAPS = Clinician-Administered PTSD Scale for Children and Adolescents; PTSD = post-traumatic stress disorder.

symptom severity was not related to trust or trust reciprocity in the current study; rather, general anxiety symptoms were related to trust reciprocity. Indicative of previous conceptualizations where PTSD is not the sole psychopathology to emerge in the aftermath of trauma exposure (Auxéméry, 2018), these findings suggest that social deficits following IV can be understood in the context of global anxious symptomatology, as opposed to utilizing PTSD-exclusive interpretations.

The present findings are in contrast to prior research that found decreased reciprocity alone among adults with social anxiety (Anderl et al., 2018) and borderline personality disorder (King-Casas et al., 2008), and even IV-related PTSD (Bell et al., 2019), indicating there is a unique aspect of trust behaviors among IV-exposed adolescents, particularly those with anxiety symptomatology, within our sample. Previous findings have indicated that trust reciprocity decreases with age throughout adolescence as a form of prosocial development (van de Groep, Meuwese, Zanolie, Güroğlu, & Crone, 2018), and prosocial behaviors such as perspective taking and social decision-making improve with age (Güroğlu et al., 2014; Tamnes et al., 2018). Furthermore, the ability to update expectations through social decision-making has been implicated to be affected in trauma-exposed individuals. In both adult and adolescent IV-exposed populations, trust-related social learning is impaired in brain regions associated with threat recognition and attention, which may lead to decreased use of social information during decision-making (Cisler et al., 2015; Lenow et al., 2014). Notably, IV-exposed adolescents seemed particularly unresponsive to trust violations (Lenow et al., 2014). This research provides potential explanations for the observed increased reciprocity, such that decreased encoding of social prediction errors could lead to misreading social cues and thereby excessive reciprocation. Misread social cues could also explain other deficits in IV victims, including less adaptive risk recognition and assertiveness in social situations (Walsh et al., 2012). The ability to accurately recognize potentially dangerous social cues can be affected by dysregulated physiological or emotional reactivity (Noll & Grych, 2011), which has been heavily implicated in post-traumatic psychopathology (Chang et al., 2018; Ehring & Quack, 2010; Heleniak et al., 2016; Villalta et al., 2018). Compared with trauma-exposed healthy controls, individuals with PTSD tend to have higher trustworthiness perceptions of others, suggesting that decreased trust tendencies in IV-exposed populations may be adaptive, and that anxious psychopathology within this population leads to maladaptive trust behaviors (Fertuck et al., 2016). Overall, the present data along with these past findings call for further exploration of social trust behaviors and the role of psychopathology on maladaptive trust among trauma-exposed adolescents.

Methodological limitations should be taken into account when interpreting the present findings. First, the grouping by anxiety diagnoses resulted in small sample sizes and an underpowered analysis, particularly for the IV^{ANX-} group. To address this, the caregiver-reported anxiety symptoms were used as a corroborative continuous analysis; however, in the finding regarding trustful investment, there was a significant effect of anxiety diagnosis while there was no effect of caregiver-reported anxiety symptoms on investment behavior. Future studies would benefit from including larger trauma-exposed control samples to increase power and better explain the role of anxious psychopathology following trauma on trust. Second, our findings may be specific to adolescent female victims of IV. Rates of PTSD following IV are nearly 3 times higher than non-interpersonal trauma (Alisic et al., 2014) which may indicate a unique relationship between the interpersonal aspect of IV and the interpersonal nature of the Trust Game. Individuals with histories of combat trauma, witnessed violence, accident trauma, or natural disasters may demonstrate different trust behaviors compared with individuals with histories of IV (Alisic et al., 2014; Williams et al., 2014). Adolescents with anxiety disorders without a history of IV exposure may also have altered trust behaviors, which has been implicated in adult samples (Anderl et al., 2018). Future studies should include trauma-naïve controls with anxiety to assess whether these findings are specific to anxiety in the presence of IV exposure or generalizable to anxiety. In addition, there is a wealth of evidence to suggest that sex and developmental differences impact prosocial behavior, social decision-making, and behavior during the Trust Game (Alisic et al., 2014; Fett et al., 2014; Lemmers-Jansen et al., 2017; van den Bos et al., 2011). Future investigations should analyze social behaviors among adolescents with varying trauma exposure types to elucidate if this finding is specific to IV in females. Furthermore, the computerized confederate in the present study was presented to all participants as an adolescent girl, and manipulation of the confederate's gender or age could play an interesting role on IV-exposed adolescents. Despite limitations, the present study is the first investigation of the behavioral changes in social trust and reciprocity in IV-exposed adolescents with and without current psychopathology. These results indicate the need for increased focus on the impact of trauma on social trust and reciprocity among adolescents, and as past research has shown (Williams et al., 2014), treatment and intervention strategies can be utilized to improve traumatized individuals' social engagement through the implementation of trauma-focused treatments for IV-exposed youth.

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